

Qualification Pack



HYDROPONICS PRODUCER

QP Code: ASP/AGR/Q0101

Version: 1.0

NSQF Level: 4

Additional Skill Acquisition Programme Kerala || Additional Skill Acquisition Programme Kerala, 3rd
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Qualification Pack

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ASP/AGR/Q0101: HYDROPONICS PRODUCER

Brief Job Description

Hydroponics producers are individuals who specialize in growing plants using the hydroponic method. Hydroponics is a soilless growing technique that involves cultivating plants in a nutrient-rich water solution.

Personal Attributes

Hydroponics Farm Entrepreneurship and managerial skills

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [ASP/AGR/N0101: BASICS OF HYDROPONICS](#)
2. [ASP/AGR/N0102: NUTRIENT MANAGEMENT](#)
3. [ASP/AGR/N0103: DESIGN AND CONSTRUCTION OF COMMONLY USED HYDROPONIC SYSTEMS](#)
4. [ASP/AGR/N0104: CROP MANAGEMENT](#)
5. [ASP/AGR/N0105: HYDROPONICS UNDER PROTECTED STRUCTURES](#)
6. [ASP/AGR/N0106: Operationalizing a Hydroponics farm](#)
7. [ASP/AGR/N0107: Management of a commercial Hydroponics farm](#)
8. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Agriculture
Sub-Sector	Agriculture Crop Production
Occupation	Hydroponics Producer, Hydroponics Producer
Country	India
NSQF Level	4

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Credits	16
Aligned to NCO/ISCO/ISIC Code	6113.0301,6113.9900
Minimum Educational Qualification & Experience	12th Class/Diploma with NA of experience OR Completed 2nd year of the 3-year diploma after 10 OR 11th Class with 1 Year of experience OR 10th Class (with two years of any combination of NTC/NAC/CITS or Equivalent) with NA of experience OR 8th Class pass (plus 2-year NTC plus 1-Year NAC plus 1-Year CITS) with NA of experience OR 10th Class () with 3 Years of experience relevant experience in on-farm work OR Previous relevant Qualification of NSQF Level (3.5) with 1.5 years of experience OR Previous relevant Qualification of NSQF Level (LEVEL 3) with 3 Years of experience relevant experience in on-farm work
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	31/01/2027
NSQC Approval Date	31/01/2024
Version	1.0
Reference code on NQR	QG-04-AG-01972-2024-V1-ASAP
NQR Version	1

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ASP/AGR/N0101: BASICS OF HYDROPONICS

Description

The candidate will obtain knowledge on hydroponics, conventional farming and plant growth requirements.

Scope

The scope covers the following :

- The candidate will obtain knowledge on hydroponics, conventional farming and plant growth requirements.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Evaluate the basic understanding of hydroponics gardening
- PC2.** Assess the knowledge on the features and advantages of soilless cultivation
- PC3.** Evaluate the knowledge of the various environmental factors affecting the plant growth
- PC4.** Knowledge about the importance of plant nutrition and essential nutrition requirements for healthy plant growth.
- PC5.** Assess the ability to differentiate between different hydroponics systems
- PC6.** Knowledge on concept and design of Hydroponics systems
- PC7.** Evaluate the awareness on site considerations for setting up Hydroponics systems
- PC8.** Evaluate knowledge on the growing media
- PC9.** Assess the basic knowledge on various pumping and electrical systems

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** BASICS OF HYDROPONICS FARMING

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	28	10	-	14
PC1. Evaluate the basic understanding of hydroponics gardening	2	-	-	1
PC2. Assess the knowledge on the features and advantages of soilless cultivation	3	-	-	1
PC3. Evaluate the knowledge of the various environmental factors affecting the plant growth	5	-	-	-
PC4. Knowledge about the importance of plant nutrition and essential nutrition requirements for healthy plant growth.	5	3	-	3
PC5. Assess the ability to differentiate between different hydroponics systems	2	4	-	2
PC6. Knowledge on concept and design of Hydroponics systems	3	-	-	2
PC7. Evaluate the awareness on site considerations for setting up Hydroponics systems	2	-	-	3
PC8. Evaluate knowledge on the growing media	4	3	-	-
PC9. Assess the basic knowledge on various pumping and electrical systems	2	-	-	2
NOS Total	28	10	-	14

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Not
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National Occupational Standards (NOS) Parameters

NOS Code	ASP/AGR/N0101
NOS Name	BASICS OF HYDROPONICS
Sector	Agriculture
Sub-Sector	
Occupation	Hydroponics Producer
NSQF Level	4
Credits	1
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	31/01/2027
NSQC Clearance Date	31/01/2024

Qualification Pack

ASP/AGR/N0102: NUTRIENT MANAGEMENT

Description

The candidate will obtain knowledge on the requirements and methodology of hydroponic nutrient management.

Scope

The scope covers the following :

- Understanding Nutrient Requirements.
- Nutrient Solution Formulation.
- Troubleshooting and Problem solving regarding nutrient management.
- Familiarize with different hydroponic nutrient delivery systems.
- Understanding the significance of pH in hydroponic nutrient management.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Basic Understanding of various nutrients and their effect on plants

PC2. Ability to select the right component salts in proper ratio

PC3. Knowledge about TDS/EC and its significance

PC4. Knowledge about Ph and its effect on plant growth

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. BASIC KNOWLEDGE

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	15	30	-	12
PC1. Basic Understanding of various nutrients and their effect on plants	2	7	-	-
PC2. Ability to select the right component salts in proper ratio	7	12	-	5
PC3. Knowledge about TDS/EC and its significance	3	4	-	3
PC4. Knowledge about Ph and its effect on plant growth	3	7	-	4
NOS Total	15	30	-	12

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National Occupational Standards (NOS) Parameters

NOS Code	ASP/AGR/N0102
NOS Name	NUTRIENT MANAGEMENT
Sector	Agriculture
Sub-Sector	
Occupation	Hydroponics Producer
NSQF Level	4
Credits	1
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	31/01/2027
NSQC Clearance Date	31/01/2024

Qualification Pack

ASP/AGR/N0103: DESIGN AND CONSTRUCTION OF COMMONLY USED HYDROPONIC SYSTEMS

Description

The candidate will obtain knowledge on designing, construction, maintenance and operation of hydroponics systems.

Scope

The scope covers the following :

- This module equips learners with the knowledge and skills to design and build functional hydroponic systems for plant cultivation.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Knowledge on differentiation and features of NFT/DFT
- PC2.** Evaluate the knowledge on Dutch bucket system
- PC3.** Evaluate the knowledge on Drip system

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	22	-	-	8
PC1. Knowledge on differentiation and features of NFT/DFT	6	-	-	4
PC2. Evaluate the knowledge on Dutch bucket system	8	-	-	2
PC3. Evaluate the knowledge on Drip system	8	-	-	2
NOS Total	22	-	-	8

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National Occupational Standards (NOS) Parameters

NOS Code	ASP/AGR/N0103
NOS Name	DESIGN AND CONSTRUCTION OF COMMONLY USED HYDROPONIC SYSTEMS
Sector	Agriculture
Sub-Sector	
Occupation	Hydroponics Producer
NSQF Level	4
Credits	1
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	31/01/2027
NSQC Clearance Date	31/01/2024

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ASP/AGR/N0104: CROP MANAGEMENT

Description

The Candidate will obtain knowledge on the best practices in crop management.

Scope

The scope covers the following :

- This NOS equips participants with the knowledge and skills to effectively manage hydroponic crop production.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Evaluate knowledge about crop selection.
- PC2.** Assess knowledge on seeding production.
- PC3.** Evaluate knowledge on Training/Pruning
- PC4.** Evaluate awareness about nutrient management
- PC5.** Evaluate awareness about best practices for plant protection
- PC6.** Assess knowledge on physiological disorders
- PC7.** Assess knowledge on harvest and post harvest operations

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	40	54	-	20
PC1. Evaluate knowledge about crop selection.	5	10	-	3
PC2. Assess knowledge on seeding production.	6	8	-	2
PC3. Evaluate knowledge on Training/Pruning	5	7	-	5
PC4. Evaluate awareness about nutrient management	6	10	-	3
PC5. Evaluate awareness about best practices for plant protection	5	6	-	2
PC6. Assess knowledge on physiological disorders	4	3	-	1
PC7. Assess knowledge on harvest and post harvest operations	9	10	-	4
NOS Total	40	54	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	ASP/AGR/N0104
NOS Name	CROP MANAGEMENT
Sector	Agriculture
Sub-Sector	
Occupation	Hydroponics Producer
NSQF Level	4
Credits	6
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	31/01/2027
NSQC Clearance Date	31/01/2024

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ASP/AGR/N0105: HYDROPONICS UNDER PROTECTED STRUCTURES

Description

The candidate will obtain knowledge on different protected structures and equipment for controlled environment agriculture and learn about hi-tech Hydroponics.

Scope

The scope covers the following :

- The candidate will gain expertise in controlled environment agriculture (CEA) through learning about various protected structures, their associated equipment, and advanced hydroponic systems.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Evaluate the understanding of Polyhouse, Rain Shelter , Nethouse

PC2. Knowledge on Indoor Farms - Vertical Farming

PC3. Awareness about New Technologies in Hydroponics with Controlled Environment Agriculture

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	20	12	-	25
PC1. Evaluate the understanding of Polyhouse, Rain Shelter , Nethouse	8	7	-	10
PC2. Knowledge on Indoor Farms - Vertical Farming	4	-	-	5
PC3. Awareness about New Technologies in Hydroponics with Controlled Environment Agriculture	8	5	-	10
NOS Total	20	12	-	25

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National Occupational Standards (NOS) Parameters

NOS Code	ASP/AGR/N0105
NOS Name	HYDROPONICS UNDER PROTECTED STRUCTURES
Sector	Agriculture
Sub-Sector	
Occupation	Hydroponics Producer
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	31/01/2027
NSQC Clearance Date	31/01/2024

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ASP/AGR/N0106: Operationalizing a Hydroponics farm

Description

The candidate will acquire the essential knowledge for starting an enterprise.

Scope

The scope covers the following :

- The candidate will acquire the essential knowledge for starting an enterprise.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Assess the ability to analyze and make decisions based on market trends

PC2. Assess knowledge about methods for effective resource Utilization

PC3. Assess ability to formulate apt marketing strategy

PC4. Evaluate knowledge on revenue model creation and Management

PC5. Assess awareness about legal compliances essential for the Business

PC6. Check the basic understanding of software tools required for the business

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	33	44	-	23
PC1. Assess the ability to analyze and make decisions based on market trends	5	-	-	5
PC2. Assess knowledge about methods for effective resource Utilization	8	12	-	5
PC3. Asses ability to formulate apt marketing strategy	5	10	-	-
PC4. Evaluate knowledge on revenue model creation and Management	5	7	-	3
PC5. Assess awareness about legal compliances essential for the Business	10	-	-	5
PC6. Check the basic understanding of software tools required for the business	-	15	-	5
NOS Total	33	44	-	23

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National Occupational Standards (NOS) Parameters

NOS Code	ASP/AGR/N0106
NOS Name	Operationalizing a Hydroponics farm
Sector	Agriculture
Sub-Sector	
Occupation	Hydroponics Producer
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	31/01/2027
NSQC Clearance Date	31/01/2024

Qualification Pack

ASP/AGR/N0107: Management of a commercial Hydroponics farm

Description

The candidate will be equipped with all the necessary capabilities to manage an enterprise based on hydroponics.

Scope

The scope covers the following :

- The candidate will be equipped with all the necessary capabilities to manage an enterprise based on hydroponics.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. To evaluate the understanding on developing Business

PC2. To assess awareness about various support schemes

PC3. Asses ability to prepare project proposals

PC4. Ability to perform the day to day activities of a farm

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	20	35	-	15
PC1. To evaluate the understanding on developing Business	5	10	-	2
PC2. To assess awareness about various support schemes	5	-	-	5
PC3. Asses ability to prepare project proposals	5	10	-	5
PC4. Ability to perform the day to day activities of a farm	5	15	-	3
NOS Total	20	35	-	15

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National Occupational Standards (NOS) Parameters

NOS Code	ASP/AGR/N0107
NOS Name	Management of a commercial Hydroponics farm
Sector	Agriculture
Sub-Sector	
Occupation	Hydroponics Producer
NSQF Level	4
Credits	1
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	31/01/2027
NSQC Clearance Date	31/01/2024

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DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

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- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

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PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings

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- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

ASAP Kerala follows an assessment framework which provides weightage for all the activities connected with skilling in which students get involved during the training program. The components of assessment include Attendance, Assignments, Internal Assessments and Final Assessment. Of these, Attendance and Internal Assessment come under Continuous and Comprehensive Evaluation (CCE). All Assessments with regard to the academic status of the student should be reflected as marks and overall assessment by awarding grades.

Attendance Criteria

Skill based sessions and training are delivered through lectures, discussions, demonstrations, and experiments ensuring student participation during daily learning activities. So those who miss such sessions will not be able to compensate for the loss. Regular student participation in daily classroom activities plays a significant role in student's success. For each course and batch, the student's attendance will be marked daily, and marks will be awarded accordingly on course completion before final assessment. Students whose attendance fall below 70% will not be eligible for final assessment and course completion certificate.

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Internal Assessment

There are two internal tests, one after completing 50% of the total training duration and the other, at the end of the training. These tests are conducted by the trainer based on the topics covered till then. In each test, 60% of the questions will be objective type and the rest 40% will be descriptive questions regarding procedure/strategies/process etc.

Assignments

Students have to submit two assignments, the first after completion of 25% of the total training duration and the second assignment after that of 75% of training duration. Both assignments shall be evaluated and assigned a score. The scores should be marked on the student's assignment sheets by the Trainer after evaluation.

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Final Assessment

1. Assessment System Overview

- Assessment division will be responsible for conducting the final assessment for the students who have completed the training with a minimum of 70% attendance.

- Question bank shall be prepared in advance by the division with the help of Subject matter experts. 2. Testing Environment

- Prior intimation will be provided to the students regarding the assessment.

- Both theoretical knowledge and practical knowledge will be assessed. Tool and equipments for practical session will be provided to the students at assessment venue.

3. Assessment Quality assurance levels/framework

- Questions papers are created by Subject matter experts considering the levels and assessment criteria.

- Assessor from the authorized Assessment agency will monitor and ensure transparency of the assessment and would support

Minimum Aggregate Passing % at QP Level : 50

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Qualification Pack

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
ASP/AGR/N0101.BASICS OF HYDROPONICS	28	10	0	14	52	5
ASP/AGR/N0102.NUTRIENT MANAGEMENT	15	30	0	12	57	5
ASP/AGR/N0103.DESIGN AND CONSTRUCTION OF COMMONLY USED HYDROPONIC SYSTEMS	22	0	0	8	30	5
ASP/AGR/N0104.CROP MANAGEMENT	40	54	0	20	114	25
ASP/AGR/N0105.HYDROPONICS UNDER PROTECTED STRUCTURES	20	12	0	25	57	10
ASP/AGR/N0106.Operationalizing a Hydroponics farm	33	44	0	23	100	20
ASP/AGR/N0107.Management of a commercial Hydroponics farm	20	35	0	15	70	20
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	10
Total	198	215	-	117	530	100

Logo
Not
Available



Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Qualification Pack

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.